Beyond League Position

Points, Points-Per-Game, and the Search for the Right Outcome Measure

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Why Look Beyond League Position?

Follow-up question: does the money–success finding survive if we change how we measure "success" — or is it an artifact of using final league position?

Two problems with the outcomes used so far:

- **League position** is ordinal and bounded (1–20): easy to compare across leagues, but blind to *margin* — finishing 1st by 1 point looks identical to finishing 1st by 20.
- **Total points** are *not* directly comparable across leagues: the Bundesliga plays 34 matches per season, the other four leagues play 38. A team on 70 points after 34 matches is dominating far more than one on 70 points after 38.

The professor's suggestion: normalise by matches played — Points Per Game (PPG) — a genuinely cross-league-comparable outcome.

This deck re-runs the full analysis three times

— Position, Points, PPG — to see whether the core finding changes.

One Correction Before Extending the Model

What we found: SPAL (Serie A, 2017–18) never matched a name during the original ESPN/Transfermarkt merge, leaving its `Market_Value` and `Financial_Rank` missing — the **only** gap in 976 team-seasons.

What we did: inserted SPAL's true 2017–18 squad value (€63.55M, Transfermarkt) and recomputed Financial Rank for that one league-season.

Independent confirmation: SPAL lands at **FR 16** — exactly the numbering gap that already existed in the untouched data (ranks jumped 15 $\rightarrow$ 17 before the fix).

Before the fix, the inflation-adjustment check (Slide 9) showed 4 apparent rank changes. All four were this one data gap, not inflation. After the fix: 0.

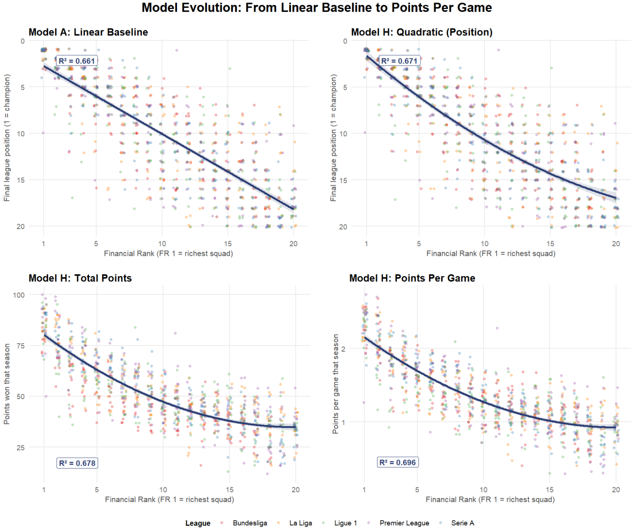
Every result in this deck — and the corresponding update to the main presentation — uses the corrected dataset (`master_dataset_corrected.csv`).

Four Specifications, One Predictor: Financial Rank

Model	Outcome	R^2	Adj. R^2	AIC	FR coef.	FR^2 coef.
A: Linear baseline	League Position	0.661	0.660	5104.4	+0.813	—
H: Quadratic	League Position	0.671	0.671	5075.1	+1.231	−0.020
H: Quadratic	Total Points	0.678	0.677	7234.6	−5.047	+0.127
H: Quadratic	**Points Per Game**	**0.696**	**0.695**	**100.1**	**−0.134**	**+0.003**

Same predictor, four scoring systems. The bottom row — the professor's suggested PPG specification — achieves the highest R^2 of any model tried in this project ($N = 976$ throughout).

Four Panels, Side by Side



What the Four Panels Show

Every panel shows the same downward-bending curve: richer squads (low FR) outperform, poorer squads (high FR) underperform, and the gap narrows as FR grows — diminishing returns hold under every outcome measure tested, from the simplest linear baseline to points per game.

Points Per Game: Why It's the Fairest Comparison

Definition

$$\text{PPG} = \frac{\text{Points}}{\text{Matches Played}}$$

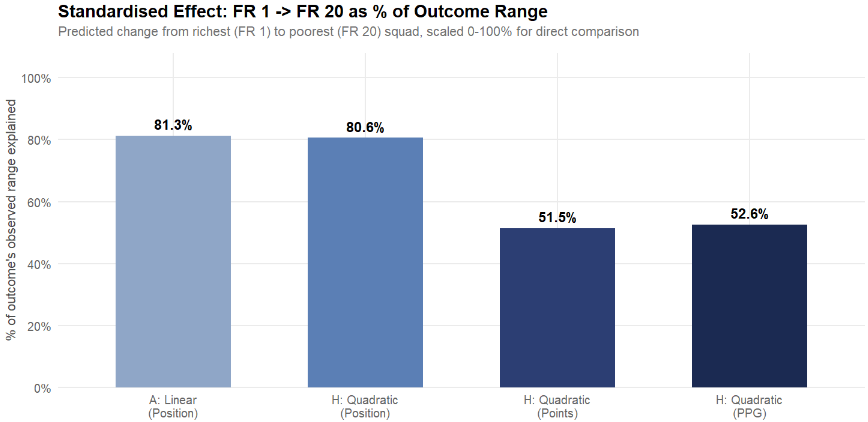
Matches played: **34** for the Bundesliga, **38** for La Liga, Ligue 1, Premier League, and Serie A.

- Removes the schedule-length distortion baked into raw *Points*.
- Puts every club, in every league, in every season, on the **same per-match scale**.

Outcome	R^2
League Position	0.671
Total Points	0.678
Points Per Game	**0.696**

PPG doesn't just satisfy the professor's request — it produces the tightest fit of all three outcome measures.

Standardised Effect Comparison: Reading the Numbers Correctly

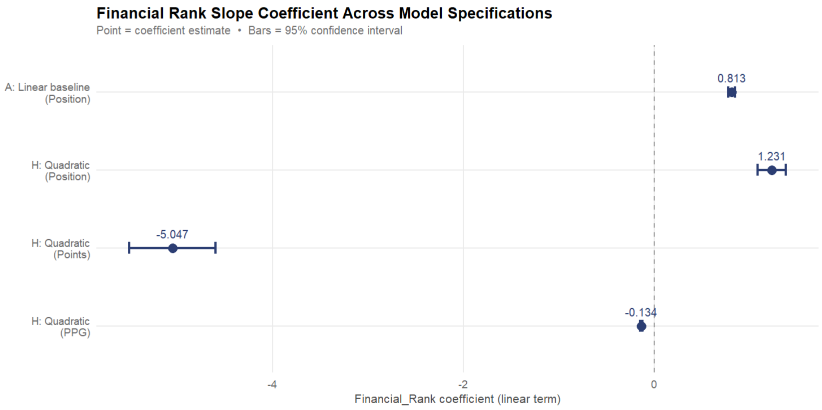


Interpreting the Standardised Effect Chart

- Chart shows: predicted change from FR 1 to FR 20, as a % of that outcome's *observed range*.
- Position models: $\approx$ 81 %. Points/PPG models: $\approx$ 51–53 %.
- **Why the gap?** League position is bounded 1–20 *by construction* — almost the entire scale is mechanically "used." Points and PPG reflect genuine match variation, not a fixed scale.

This is not evidence that PPG predicts "less well." R^2 (Slide 4) is the correct measure of fit — and PPG scores highest there. This chart measures something different: how much of the observed spread the FR effect spans.

Coefficient Evolution Across Specifications



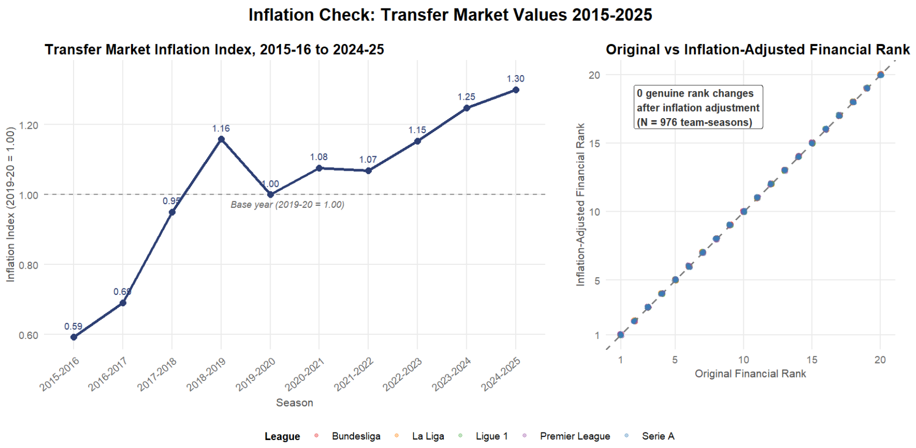
The Sign Flip Is Definitional, Not a Contradiction

Statistical: The Financial_Rank coefficient is **positive** for Position ($+0.81$, $+1.23$) and **negative** for Points and PPG (-5.05, -0.13).

Plain English: Higher FR always means a worse outcome — it just points in opposite arithmetic directions depending on whether "worse" means a bigger position number or a smaller points total.

Whichever way the sign points, the message is identical: moving from the richest squad (FR 1) to the poorest (FR 20) costs a club dearly, on every scale tested.

Does Financial Rank Itself Survive Ten Years of Inflation?

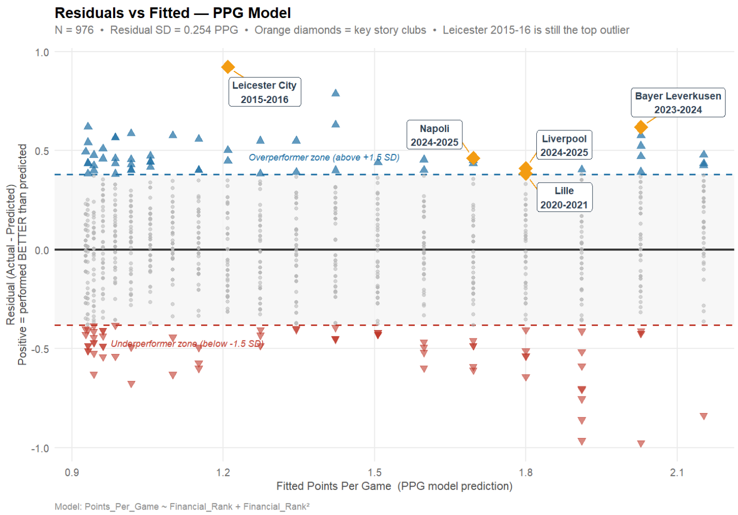


Why This Matters for Every Model in This Deck

- Average squad value roughly **doubled** 2015–16 to 2024–25 (index $0.59 \rightarrow 1.30$, $2019–20 = 1.00$).
- But **Financial_Rank** is a *within-season* rank, not a raw euro figure — dividing every club in a league-season by the same constant cannot change their relative order.
- After the SPAL fix: **0 genuine rank changes** in all 976 team-seasons, confirmed individually in *every* one of the five leagues.

Every model in this deck — Position, Points, PPG — is fed the same season-stable predictor. Inflation cannot be driving any of these results.

PPG Residuals: Does Leicester Still Win?



Leicester City 2015–16: Still the Greatest Outlier

PPG Model Diagnostics	
Residual SD	0.254 PPG
± 1.5 SD threshold	± 0.381 PPG
Overperformers	56 teams
Underperformers	71 teams

Leicester City 2015–16: residual $= +0.923$ PPG — the single largest residual of all 976 team-seasons, rank #1. Even under the professor's preferred specification, Leicester's title remains the greatest financial upset in the dataset.

Two Honest Caveats About PPG

COVID-Shortened Ligue 1, 2019–20

Season stopped after **27** of the assumed 38 matches. PPG still divides by 38 (per the original specification), so that season's Ligue 1 PPG values are biased **downward** relative to completed seasons. Flagged, not corrected.

Ligue 1's 2023–24 Format Change

Ligue 1 moved to an 18-club, **34**-match format from 2023–24 onward — the same 34-vs-38 divisor mismatch affects those two seasons.

Statistical: 3 of the 50 league-seasons in the dataset deviate from the assumed 38-match schedule (all Ligue 1).

Plain English: A handful of Ligue 1 PPG values should be read with this caveat in mind — no other league or season is affected.

Conclusion: Same Story, Sharper Lens

1. Financial Rank predicts success under every outcome measure tried.

$R^2 = 0.671$ (Position), 0.678 (Points), **0.696** (PPG) — all $N = 976$, all $p < 0.001$.

2. PPG is the most defensible outcome measure of the three.

The only measure directly comparable across leagues with unequal season lengths — and it fits best.

3. Leicester City 2015–16 is not an artifact of choosing "league position."

It is the single largest overperformance under points **and** under PPG — the anomaly survives every outcome measure.

4. One data gap, now closed.

A single missing team-season (SPAL, 2017–18) was the entire source of an earlier inflation-adjustment artifact — now resolved, with zero genuine rank changes remaining.

Whichever way you count success — position, points, or points per game — money predicts it. The professor's suggested PPG metric doesn't just confirm the original finding: it is the tightest fit of them all.

Thank You — Questions?

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$$R^2_{\textbf{PPG}} = 0.696 \quad \textbf{vs.}\ 0.671\ \textbf{(Position)} \qquad N = 976$$

Three outcome measures · One predictor · One answer: **money predicts success, however you score it**

Key Numbers to Remember

- **0.696**: highest R^2 of any model, using PPG
- **0** genuine rank changes after the SPAL fix (was 4)
- **34 vs 38**: matches played, Bundesliga vs. the other four leagues
- **+0.923 PPG**: Leicester City 2015–16's residual — still #1 of 976

The Anomaly That Won't Go Away

Leicester City 2015–16

Position residual: -10.5 positions (-3.2 SD)
Points residual: largest in dataset
PPG residual: $+0.923$ ($+3.6$ SD), **rank #1**

Three outcome measures, three confirmations: one genuine miracle.

Data: ESPN FC + Transfermarkt (corrected) **Period:** 2015–16